

Supplementary Material: Adaptive Communication Topology Learning (AC-DSGF)

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S1. NETWORK ARCHITECTURE (SUMMARY)

AC-DSGF wraps a spatial-temporal graph encoder with a differentiable gate head. Node embedding dimension $d_h=128$; spatial layers use communication-aware sparse attention (2 layers, 4 heads); residual guidance dimension $d_\Phi=6$. Gates $g_{ij} = \sigma(s_{ij}) \cdot A_{ij}$ are restricted to the radius support A_{ij} ($R_c=0.5$). Soft training cost $C_s = \sum g_{ij}$ (SCAM) is penalized by λ_c in $\mathbb{E}[R - \lambda_c C_s]$. Hard deployment uses thresholding or per-agent Top- K .

S2. HYPERPARAMETERS (FROZEN V1)

Item	Value
Agents N (main)	16
Episode horizon	128
MAPPO / CTDE	as in baseline configs
λ_c (main checkpoint)	10^{-3}
Comm. radius R_c	0.5
Success threshold	0.3
Eval protocol	TorchRL step_mdp

Full YAML: `configs/ac_dsgf/ac_dsgf_16uav.yaml`.

S3. PROPOSITION 1 DERIVATION (DEGREE CONSTRAINT)

Assume $|E_t(i)| \leq K$ for each agent. Then $|E_t| \leq NK$. The complete digraph has $|E_{\text{full}}| = N(N-1)$, hence

$$\eta_N = \frac{|E_t|}{|E_{\text{full}}|} \leq \frac{K}{N-1} = \mathcal{O}\left(\frac{1}{N}\right).$$

This is a *scaling interpretation under bounded neighborhood*, not a learning-convergence claim and not an equality $|E_t| = KN$.

S4. WHAT IS *not* IN THIS APPENDIX

No additional high-Success leaderboards; no SwarmOS/PX4 engineering claims; no multi- λ retrain curves claimed as main-text results.